A Ship Honeynet to Gather Cyber Threat Intelligence for the Maritime Sector

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***Abstract*— This article discusses the ongoing design and development of a ship Honeynet for the purpose of gathering Cyber Threat Intelligence (CTI) to assist the Maritime Sector to be more resilient in the face of increasing cyber-attacks. The critical and fragile nature of the Global Maritime Transportation System (GMTS) has been clearly shown during the COVID-19 Pandemic and in the ongoing impact of the Russian invasion of Ukraine. Unfortunately, at the same time the industry is highly vulnerable to cyber-attack, given ship fleets are aging and their technology is aging with them while at the same time, there is a low level of cyber security maturity within the industry. This article describes the overall research project aiming to gather information on current cyber-attacks on vessels using a Honeynet to develop CTI. This CTI will inform cyber security and maritime professionals to better respond to those threats. Additionally, the project will provide data for a broad range of future research. Honeypots are Internet systems deployed for the sole purpose of being compromised to observe adversaries. Networks of Honeypots are termed Honeynets. Honeypots/Honeynets have existed for more than 30 years however simulating all the different systems of a ship in a Honeynet to research cyber-attacks on maritime is a novel concept. A real ship consists of multiple digital systems which perform and assist in a range of necessary functions such as navigation, communication, safety, propulsion, cargo management, other IT standard office activities, and numerous other purposes. This article will explain the concept of Honeynets, its history, and a ship Honeynet in particular, as well as its design considerations, existing models, and benefits. This article will also look in detail at the data flows for the ship Honeynet and in a new concept how it can be integrated using MITRE Corporation's STIX and TAXII methodologies for CTI purposes.**

Keywords—Cybersecurity, Maritime Security, Cyber-Physical Security, Vessel, Honeypot, Honeynet, Cyber Deception.

I. Introduction

This article will discuss the ongoing design and development of a ship Honeynet to gather information on current cyber-attacks on vessels and develop CTI [1]. The criticality and fragility of the GMTS is a significant international problem impacting both the global economy and global security. The GMTS is a system of systems and includes not just vessels but also waterways, ports, and land-side connections, moving people and goods to and from the water. The role of GMTS in the global economy is significant with over 80-90% of the world's cargo transported via the GMTS [2] and it represents 70% of global trade by value [3]. Unfortunately, at the same time, the industry is highly vulnerable to cyber-attack, given ship fleets are aging and their technology is aging with them and there is a low level of cyber security maturity within the industry [4]. Some 38% of oil tankers and 59% of general cargo ships are more than twenty years [5]. Supply chains themselves are increasingly vulnerable to cyber-attacks. This is particularly stark in recent years, "…European sources estimated a 400% growth in supply chain cyberattacks in 2021 compared to 2020" [6]. GMTS is a fundamental part of global supply chains and will be increasingly targeted by cyber threat actors. Since 2018, state-sponsored threat actors from China and Iran (amongst others) have specifically targeted the maritime industry with malware that is designed to compromise technology only used in shipping [7]. A recently published online database by researchers from NHL Stenden University of Applied Sciences exposes a large list of incidents that occurred in the worldwide maritime industry [8]. The database contains over 160 incidents gathered from various open sources.

Honeypots are Internet systems deployed for the sole purpose of being compromised to observe cyber adversaries. Networks of Honeypots are termed Honeynets and, like network telescopes, are typically deployed on an otherwise unused address space [9]. Honeypots/Honeynets have existed for more than 30 years however simulating all the different systems of a ship in a Honeynet to research cyber-attacks on maritime is a novel concept. A real ship consists of multiple digital systems which perform and assist in a range of necessary functions such as navigation, communication, safety, propulsion, cargo management, other IT standard office activities, and numerous other purposes. The ship Honeynet needs to simulate many of these things.

Part of the process is to make the Honeynet appear realistic and interesting to potential cyber-attackers and the article identifies a number of features that would make the Honeynet

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more realistic and thus more likely to attract and engage attackers. The ship Honeynet is going to use a technique proposed by researchers [10] called intelligent interaction to assist in this. The paper also discusses methods to use captured interactions with those attackers including connection details, commands executed, files dropped, and other relevant activity, and use this data to produce CTI.

The structure of this article is firstly the background of Honeynets, etc., followed by a description of the cyber threats to the maritime sector, design considerations for a ship Honeynet, how to leverage existing techniques to enhance CTI gained, and, finally, the conclusions and future research.

II. Background of Deception, Honeypots, and Honeynets

Honeypots and the use of deception against cyber attackers date back over 30 years. Clifford Stoll, a research astronomer in his well-known book, The Cuckoo's Egg, described when working as a part-time system administrator at Lawrence Berkeley National Lab in the USA his methods to uncover cyber-attackers who had penetrated his system [11]. This early Honeypot was born of his scientific research approach to observe his cyber-attackers and get them to reveal more of themselves, "Do research...OK, I'll watch the guy and call it science" [12]. In 1999 the Honeynet Project was formed with 30 members from the, at that stage, small cyber security community. Amongst that group of 30 was Lance Spitzner and he described a Honeypot as, "A … security resource whose value lies in being probed, attacked, or compromise" [13].

Some common deployment strategies for Honeypots were described by Scottberg et al. [14]. They include, "Sacrificial Lamb", an isolated system that has no entry point to production systems. A "Hacker Zoo", an entire subnet of Honeypots with varied platforms, services, vulnerabilities, and configurations, which are isolated from production systems. A "Minefield", a number of Honeypots placed in forefront to serve as first attack targets. A "Proximity Decoy", a Honeypots deployed in close proximity to production systems. A "Redirection Shield External", that appear on production systems through port redirection and lastly a "Deception Port", simulating services (e.g., SMTP, DNS, FTP) on production systems.

A honeynet can been seen as a powerful trap to lure attackers into there is always the risk of attackers breaking out of honeypots. According to Zhang and Vrizlynn attackers are highly motivated to detect honeypots in an early phase of their kill chain [15]. So that their efforts are not in vain and their TTPs are not disclosed.

III. Critical Cyber Threat to Maritime

As stated in the introduction the criticality and fragility of our supply chains is particularly evident within the GMTS. In a 2019 report 'Shen attack: Cyber risk in Asia Pacific ports' – produced by the University of Cambridge Centre for Risk Studies, researchers described a hypothetical cyber-attack across the Asia Pacific. The cyber-attack against 15 ports utilized malware that jumped from ships to ports. The report projected the loss could go as high as USD$110 Billion and the vast majority of that amount would not be covered by any insurance [16]. We have not yet seen a cyber-attack on this scale in the maritime sector, but we have seen numerous ports and ships impacted by various cyber-attacks including ransomware, destructive malware, and even cyber-attacks on Operational Technology (OT) [1]. The threat actors responsible for these cyber-attacks include both criminal groups and nation-state hackers. The well-documented case of Maersk in 2017 where their losses were over USD$200 million in 2017 is a noteworthy example [17].

IV. Honeypot Design and Structure

A. *Ship Technical Architecture*

Ships are a complex network of a diverse range of information technology (IT), operational technology (OT), and communication technologies used onboard. Often ships also have all these technologies directly connected to their IT networks. To avoid unnecessary complexity for this initial version of the ship Honeynet the focus is to simulate the Integrated Bridge System (IBS) of a ship. The IBS acts as the main command and control of a vessel as it interconnects various digital devices used for navigation in open seas and is also connected to other on-board systems of a vessel, e.g., navigation and control, propulsion and machinery management system, cargo management system and safety management system, core infrastructure systems, administrative and crew welfare systems, etc. [18]. The International Maritime Organization (IMO) defines an IBS as a combination of systems which are interconnected in order to allow centralized access to sensor information or command/control from workstations, with the aim of increasing safe and efficient ship management by suitably qualified personnel [19].

The main components that are part of the IBS are Automatic Identification System (AIS), Electronic Chart Display Information System (ECDIS), radar, conning display, Bridge and Watch Alarm System (BNWAS) / Bridge Alert Management System (BAMS), Voyage Data Recorder (VDR), and autopilot. Sensors like a compass, speed log, and echo sounder are also providing information to the system in the IBS [20]. There is also a Satellite terminal connected to provide the access the Internet.

B. *Making the Ship Honeynet an Attractive Target*

As the success of the ship Honeynet relies on cyber-attackers engaging with it, the design considerations need to address how to make the ship Honeynet attractive and believable to potential cyber-attackers.

If the focus is the Integrated Bridge System (IBS) then to make it believable and therefore attractive to cyber attackers a number of features need to be considered [21]:

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- Logical server location in domain/IP records.
- Logical sailing route with realistic AIS data (see below).
- Network speeds appropriate to a satellite connection (therefore slow).
- Network signs of life with realistic traffic between systems.
- A logical entry point for cyber attackers i.e., Satcom, remote access portal, etc. (see below)
- A system architecture that is appropriate to the type, size, and mission of a ship.

These features will be implemented in the proposed design. The ship Honeynet will also use a technique called intelligent interaction [10]. The goal of intelligent-interaction is to learn the 'correct' behaviors to interact with clients from zero knowledge about the ship Honeynet.

C. *The entry point for ship Honeynet*

The proposed entry point for a cyber attacker into the IBS will be the satellite terminal for the first version of the ship Honeynet. Different vulnerability reports have revealed the misconfiguration of these types of remote management terminals is common. A cyber attacker scanning the IP address will find for example an open SSH port of the satellite terminal to connect to and enumerate it. In particular the cyber-attacker can gain access to the IBS thru the commonly used Cobham (Satellite) SeaTel terminal. This type of terminal is being used as a gateway to the Internet and can be emulated in ship Honeynet. The Cobham SeaTel terminal has vulnerabilities with existing exploits regarding the injection of malicious JavaScript using the devices' TELNET built-in commands [22]. In this way, the cyber-attacker can gain access to the IBS directly from the Internet and then move around the connected ship Honeynet in a realistic fashion.

V. Cyber Threat Intelligence

CTI has a long history in the cyber security field [23]. Much of the early CTI was information that helped identify the malware of the mid-1990s. It tended to be very technical in nature and largely focused on the malicious code, its execution, infection methods, and strings and code fragments to help identify it in the wild. In the 2000s CTI started to talk about campaigns of cyber-attacks, where numerous associated events could be described. Lastly, CTI began to address threat actors themselves their Techniques, Tactics, and Procedures (TTPs), their motivation, capability, and even their identities. This threat actor-oriented CTI is where Honeynets and Honeypots are most useful. Accessing data on threat actors from the wild such as this ship Honeynet allows is of significant benefit to this type of CTI.

A. *Sharing of Cyber Threat Intelligence*

There are some issues with manual threat intelligence sharing. In a paper published by Wagner, Thomas D., Khaled Mahbub, Esther Palomar, and Ali E. Abdallah [24] they mention that current CTI sharing is heavily based on manual input and therefore is labor intensive. A large amount of current CTI sharing is conducted through e-mails, phone calls, web-community portals, shared databases, and data feeds. So, there is a real urge to adopt automation with the significant volume of internal alerts and externally received information about vulnerabilities and it is proposed to automate that sharing in the proposed ship Honeynet design.

B. *Cyber Threat Intelligence Sharing Protocols*

In the previous section sharing of automated CTI information was introduced. Different studies list Structured Threat Information eXpression (STIX) as the best candidate for sharing CTI information and the most used [24], [25]. STIX architecture provides expressive, flexible, and extensible representation language used to communicate an overall piece of threat information. The STIX architecture is capable of describing the different types of cyber threat information such as cyber observables, indicators, incidents, adversaries, tactics, techniques, procedures, exploit targets, courses of action, cyber-attack campaigns., and threat actors. STIX was recently updated to version 2.1. Trusted Automated Exchange of Intelligence Information (TAXII) is a protocol for automated sharing in real-time of cyber threat incidents in the STIX format. Krasznay and Gyebnar discuss the way to describe an incident by using the Structured Threat Information Expression (STIX), version 2.1. They used the standardized methods of STIX 2.1 to create their CTI feed. Then they are able to share information via TAXII [26]. This is the model adopted for the proposed ship Honeynet design.

C. *Honeynet Sharing Cyber Threat Intelligence.*

In a paper published by Kumar, Janet, and Eswari there is a system design architecture proposed for a multi-platform honeynet/honeypot [27]. Their proposed system design consisted of four major building blocks. The issue with this design is that sharing of the CTI information is not yet implemented. From this design, it is possible to extract the four main domains for the ship Honeynet and add an extra domain for feeding the honeynet with live data. Namely:

1) *Para-virtualization Hypervisor:*

The software enabling the para-virtualization containers is running on the base operating system by using the hardware resources of the base machine. The containers are created and make them live as instances of honeypot images. The honeypot's threat capturing sensors are compiled and enabled as an instance of the containers [27]. The hypervisor can be different systems for example VMWare, or Proxmox VE. For example, every virtualization system that is able to run and start Docker images.

2) *Honeypot images*

Honeypot images are designed in such a way that the attacker will not be able to fingerprint them and therefore they look like a real production resource. This is inspired by the game

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theory approach in the field of Artificial Intelligence, to mimic the environment as per the cyber-attacker perspective. The baseline of Honeypot containers is performed to make the balance between normal machine and hardened machine so that attacker will not be able to guess the target is a honeypot [27].

3) Analysis Engines Blocks

In this process of system design, the analysis engines are running integrated with the system. The various analysis techniques for known attack data detection is implemented in this block. Techniques such as Signature based, and Pattern Knowledge based are run as an instance to process and alert on attack data. With the detection and labelling of unknown attack patterns, the deep learning-based methods are applied such as CNN (Convolutional Neural Network) or malware classifications [27].

4) Threat Visualization and Cyber Threat Intelligence Generation

Once the data is collected from Honeypot sensors, it is analyzed through the analysis block and then there is a need to convert this raw information into actionable formats to be directly digested by a machine in the form of CTI. They suggest the cyber threat data sharing STIX 2.0 [27].

5) Live data

In the design, the authors opted for a fifth domain namely the integration of a simulated live data feed in the ship Honeynet because of its specific nature (i.e. course and location). Being a ship Honeynet this helps simulate a ship sailing in a realistic fashion. This added block can be seen in Figure 1. The block has the goal to feed the system with realistic live data for a ship. When a ship is sailing there is all kinds of information traveling over the different bus systems of the ship. The goal of this block is to make the ship Honeynet as realistic as possible. The Honeynet is provided with, for example, replayed AIS data that might represent a vessel sailing between the Harlingen and Terschelling in the Netherlands.

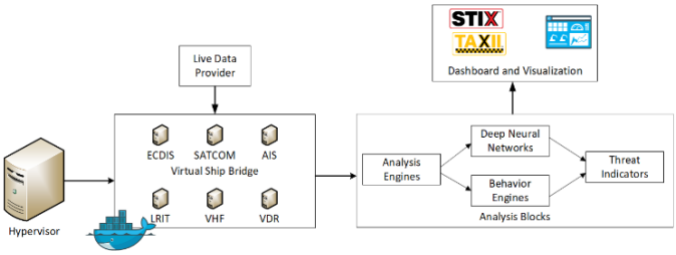

Figure 1. Five Block ship Honeynet architecture based on "System Design Architecture" [27]

D. Minimal ship honeynet architecture

A basic honeynet architecture depicts three essential elements: data control, data capture, and data collection [9]. Data control involves controlling the flow of data so that the attackers do not realize they are in a honeynet and making sure that if the Honeynet is compromised it will not be used to attack other systems. The data capture involves recording all the data regarding movements and actions within the Honeynet. The data collection involves the ability to securely transfer all the captured data to a centralized place. Figure 1 represents the ship Honeynet in five main blocks. The proposed model itself is based on the design architecture for a multi-platform Honeynet proposed by Kumar et al [27] and adapted to replicate a maritime and ship systems environment. The cyber-attacker of the ship Honeynet has only one entry point into the Honeynet. This entry is the satellite terminal within the Integrated Bridge System (IBS) as described above. The satellite terminal has numerous vulnerabilities that the cyber-attacker can leverage for access discussed further below. When the attacker gains access to the integrated bridge of the system they can only connect with systems of the IBS see Figure 2 which was proposed by undergraduate students working with the authors to develop a model [21].

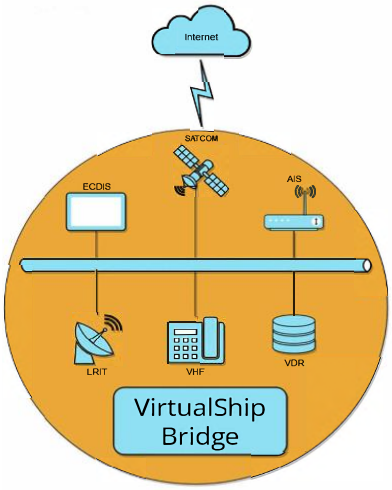

Figure 2. Minimal architectural drawing of the ship Honeynet [21].

As stated, to make the IBS more realistic to the attacker there is the Live Data Provider playing important information to the IBS like simulating Automatic Identification System (AIS) data. At this moment the system is fetching data into the IBS by running a Python script called VDRPlayer that is altered to let the ship Honeynet to seemingly sail between Harlingen and Terschelling. In the next design iteration, there is the consideration that the ship Honeynet also sends "real" data into the different online AIS viewers like AISHub, Shipfinder.co, MarineTraffic, VesselFinder and so on. AISHub provides the AIS Dispatcher which is an AIS data forwarding utility which provides a way to upload data to the different online applications.

E. Ship Honeynet data flow architecture

All honeypots collect data about the used attacks and possible dropped malware. It is proposed to use the following dataflow in the ship Honeynet. In Figure 1 the Honeynet dataflow design is illustrated. Firstly, the hypervisor block that

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enables the Honeynet to run the environment to host the Docker images. Secondly, the block that's called the Virtual Ship Bridge, which is the real goal for the cyber-attacker. Thirdly the live data provider feed sends information to the emulated IBS and the ship Honeynet captures the information from the cyber-attacker and potentially any malware or files the attacker drops. The information is then transported forthly to the analysis block and from there lastly it is sent to fifth block Dashboard and Visualization and is made available in the STIX format. With the analysis block we take care of the different steps for analysis. The way that is proposed for this project of the analyzing block is inspired by recent research by Kumar et al. [27] and also the work done by Honeynet Threat Sharing Platform [28]. The goal of the analyzing block is that the collected data sets are processed and converted into a set of feature matrixes that then can be processed by deep learning models. Also, within the analyzing block, there is a behavior component. Various analyzing techniques will be used to identify and class behavior. The dashboard and visualization block is responsible for converting the raw information into an actionable format to be directly digested by a machine in the form of CTI. The ship Honeynet proposed to use STIX format latest version 2.1. TAXII is then used for sharing via services and message exchanges. TAXII is built to support the STIX framework. It does so by an API that aligns with common sharing models. In the future, the option of the ELK stack will be investigated. The attacker might leave for example a binary payload. In a future implementation, the sample needs to be further investigated in an automated fashion. In a paper published by Kyriakou and Sklavos [29] the malware samples were automatically sent to VirusTotal where they are analyzed and successfully identified as malicious. Another option is that MITRE organization also offer Malware Attribute Enumeration and Characterization (MAEC), a very expressive malware sharing language for encoding and communicating high-fidelity information about malware based upon attributes such as behaviors, artifacts, and attack patterns. MAEC also offers the option to be integrated into STIX or used standalone [30].

Figure 1 also describes the components that will each be hosted in Docker containers as described above. A Docker container image is a lightweight, standalone, executable package of software that includes everything needed to run that application: code, runtime, system tools, system libraries, and settings [31]. In practice, a container is easy to deploy and maintain. Utilizing docker containers also provides security to prevent cyber attackers from jumping from the ship Honeynet to the host system.

VI. Conclusions

While the design, development, and operation of a ship Honeynet is a very complex project the benefit of the resulting CTI makes it a worthwhile effort. It will also produce datasets for future research benefiting not just the maritime sector but other sectors with similar cyber security challenges such as manufacturing and energy. The raw data collected includes the full packet capture of the cyber-attacker connections to the ship honeynet which include all connection information including timing and all content of the communication. This shows the source and footprint of the system connecting, commands executed, binaries dropped, etc. From this data indicators of attack (IOAs) and/or compromise (IOCs) can be drawn such as the Internet Protocol (IP) address of a suspected command and control server, a suspicious Domain Name System (DNS) domain name, a Uniform Resource Locator (URL) that references malicious content or a file hash for a malicious executable. Also, with this data, we can identify the cyber-attackers' tactics, techniques, and procedures (TTPs). The National Institute of Standards and Technology (NIST) describes TTPs as:

"The behavior of an actor. A tactic is the highest-level description of this behavior, while techniques give a more detailed description of behavior in the context of a tactic, and procedures an even lower-level, highly detailed description in the context of a technique."[32].

The project itself will evolve a series of ship Honeynets to build capability and explore different aspects of the maritime sector. The cyber ship Honeynet templates will eventually be made available to other researchers and cyber security practitioners for their use and further development. Currently, there is very limited information on the level and nature of cyber-attacks on maritime and in particular maritime vessels. For example, a publication in 2021 was only able to describe some 46 discrete cyber-attacks dating back to 2010 across the entire maritime sector [33]. This research will produce significant and up-to-date intelligence to fill this gap. In addition, the use of MITRE Corporation's STIX and TAXII methodologies will allow the gathered data to be used in cyber security tools and for various tactical and strategic intelligence purposes. As stated, the ultimate purpose of a Honeynet is to be "probed, attacked, or compromised" by cyber attackers, and by this process, we learn more of the nature of those attacks, the threat they pose, the modus operandi of those attackers including their TTPs, their motives and other relevant features of their activity. This CTI will be invaluable to the cybersecurity and maritime industry to be used for industry awareness, appropriate defensive measures and to inform future procedures and technology development. Also, it will be of significant use to researchers to create threat reports/publications, reporting any identified vulnerabilities to vendors, and creating realistic maritime cyber incident simulations.

A. *Future Research*

Beyond the planned activities to build, test, deploy the ship Honeynet and analysis the captured data a number of other further research areas exist. A future model of the ship Honeynet should represent both Information Technology (IT) and Operational Technology (OT) networks in ship environments. While many environments on land where these technologies are

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mixed are largely segmented in contrast ships due to legacy systems and more limited options due to space are often accessible to a cyber attacker on one network or via bus systems used in ship bridge technology. This means a cyber-attack can easily jump from IT to OT with all sorts of implications for crew and ship safety. The maritime industry has a lot of standards for interconnecting components and understanding these will be key to future research. The most relevant protocols are NMEA 2000, NMEA 0183, TCP/IPv6, and the latest one NMEA OneNet [34]. These protocols could be implemented as the basis of future ship Honeynet designs. Other options include building various ship Honeynets to represent various types of ships of all sizes in different global locations and involved different functions. Perhaps the most important piece of future research as mentioned above is examining the captured data from attacks. Work could include a profile of the nature of those interactions, artifacts gathered, links to known threat actors, and known toolsets. It could also assess whether it appears if cyber-attackers were fooled or simply blindly scanning. Another key research question to be explored is how useful is that data in predicting or preventing future cyber-attacks on ships. Also, the effect of running the ship Honeynet in terms of deception and disruption of the cyber attacker and the potential for greater deception and disruption if they were deployed on a large scale.

REFERENCES

[1] S. J. McCombie and J. Pijpker, "A Ship Honeynet Project to Collect Data on Cyber Threats to the Maritime Sector," presented at the CYBER 2022, The Seventh International Conference on Cyber-Technologies and Cyber-Systems, Nov. 2022, pp. 81–85.

[2] C. Bronk and P. Dewitte, "Maritime Cybersecurity: Meeting Threats to Globalization's Great Conveyor," presented at the Hawaii International Conference on System Sciences, 2020. doi: 10.24251/HICSS.2020.240.

[3] W. Loomis, V. V. Singh, D. G. C. Kessler, and D. X. Bellekens, "Raising the Colors: Signaling for Cooperation on Maritime Cybersecurity".

[4] K. Tam, K. Jones, and M. Papadaki, "Threats and Impacts in Maritime Cyber Security," *Eng. Technol. Ref.*, vol. 1, Jan. 2016, doi: 10.1049/etr.2015.0123.

[5] K. Tam and K. Jones, "Maritime cybersecurity policy: the scope and impact of evolving technology on international shipping," *J. Cyber Policy*, vol. 3, pp. 1–18, Aug. 2018, doi: 10.1080/23738871.2018.1513053.

[6] G. C. Kessler and S. D. Shepard, *Maritime cybersecurity: a guide for leaders and managers*. Gary C. Kessler and Steven D. Shepard, 2020.

[7] "Suspected Chinese Cyber Espionage Group (TEMP.Periscope) Targeting U.S. Engineering and Maritime Industries," *Mandiant*. https://www.mandiant.com/resources/blog/suspected-chinese-espionage-group-targeting-maritime-and-engineering-industries (accessed Jul. 17, 2023).

[8] "Researchers at NHL Stenden launch database that exposes cyber hacking in the worldwide maritime industry | NHL Stenden university of applied sciences," Jul. 13, 2023. https://www.nhlstenden.com/en/news-and-articles/researchers-at-nhl-stenden-launch-database-that-exposes-cyber-hacking-in-the (accessed Jul. 17, 2023).

[9] J. Franco, A. Aris, B. Canberk, and A. S. Uluagac, "A Survey of Honeypots and Honeynets for Internet of Things, Industrial Internet of Things, and Cyber-Physical Systems." arXiv, Aug. 04, 2021. Accessed: Oct. 04, 2022. [Online]. Available: http://arxiv.org/abs/2108.02287

[10] T. Luo, Z. Xu, X. Jin, Y. Jia, and X. Ouyang, "IoTCandyJar: Towards an Intelligent-Interaction Honeypot for IoT Devices," p. 11.

[11] C. Stoll, "Stalking the wily hacker," *Commun. ACM*, vol. 31, no. 5, pp. 484–497, May 1988, doi: 10.1145/42411.42412.

[12] C. Stoll and J. W. D. Connolly, "The Cuckoo's Egg: Tracking a Spy Through the Maze of Computer Espionage," *Phys. Today*, vol. 43, no. 8, pp. 75–76, Aug. 1990, doi: 10.1063/1.2810663.

[13] L. Spitzner, *Honeypots: tracking hackers*. Boston: Addison-Wesley, 2003.

[14] B. Scottberg, W. Yurcik, and D. Doss, *Internet honeypots: protection or entrapment?* 2002, p. 391. doi: 10.1109/ISTAS.2002.1013842.

[15] L. Zhang and V. L. L. Thing, "Three Decades of Deception Techniques in Active Cyber Defense -- Retrospect and Outlook," *Comput. Secur.*, vol. 106, p. 102288, Jul. 2021, doi: 10.1016/j.cose.2021.102288.

[16] J. Daffron *et al.*, "Shen attack: Cyber risk in asia pacific ports," *Cent. Risk Stud. Camb.*, 2019.

[17] L. Mathews, "NotPetya ransomware attack cost shipping giant Maersk over $200 million," *Forbes Aug*, vol. 16, 2017.

[18] M. S. K. Awan and M. A. Al Ghamdi, "Understanding the Vulnerabilities in Digital Components of an Integrated Bridge System (IBS)," *J. Mar. Sci. Eng.*, vol. 7, no. 10, Art. no. 10, Oct. 2019, doi: 10.3390/jmse7100350.

[19] *The Guidelines on Cyber Security Onboard Ships*, vol. 4. BIMCO, 2017.

[20] C. Hemminghaus, J. Bauer, and E. Padilla, "BRAT: A BRidge Attack Tool for Cyber Security Assessments of Maritime Systems," *TransNav Int. J. Mar. Navig. Saf. Sea Transp.*, vol. 15, no. 1, pp. 35–44, 2021, doi: 10.12716/1001.15.01.02.

[21] R. Rijnierse, E. Vedelaar, S. van der Veen, and S. El Farj, "Ship Honeynet Student Project," 2022.

[22] "CVE - CVE-2018-5071." https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2018-5071 (accessed Jul. 17, 2023).

[23] W. Tounsi and H. Rais, "A survey on technical threat intelligence in the age of sophisticated cyber attacks," *Comput. Secur.*, vol. 72, pp. 212–233, Jan. 2018, doi: 10.1016/j.cose.2017.09.001.

[24] T. D. Wagner, K. Mahbub, E. Palomar, and A. E. Abdallah, "Cyber threat intelligence sharing: Survey and research directions," *Comput. Secur.*, vol. 87, p. 101589, Nov. 2019, doi: 10.1016/j.cose.2019.101589.

[25] C. Sauerwein, C. Sillaber, A. Mussmann, and R. Breu, "Threat Intelligence Sharing Platforms: An Exploratory Study of Software Vendors and Research Perspectives," *Wirtsch. 2017 Proc.*, Jan. 2017, [Online]. Available: https://aisel.aisnet.org/wi2017/track08/paper/3

[26] C. Krasznay and G. Gyebnar, "Possibilities and Limitations of Cyber Threat Intelligence in Energy Systems," in *2021 13th International Conference on Cyber Conflict (CyCon)*, Tallinn, Estonia: IEEE, May 2021, pp. 171–188. doi: 10.23919/CyCon51939.2021.9468289.

[27] S. Kumar, B. Janet, and R. Eswari, "Multi Platform Honeypot for Generation of Cyber Threat Intelligence," in *2019 IEEE 9th International Conference on Advanced Computing (IACC)*, Tiruchirappalli, India: IEEE, Dec. 2019, pp. 25–29. doi: 10.1109/IACC48062.2019.8971584.

[28] "Technical Report – ISIF ASIA." https://isif.asia/collaborative-honeynet-threat-sharing-platform/ (accessed Jul. 17, 2023).

[29] A. Kyriakou and N. Sklavos, "Container-Based Honeypot Deployment for the Analysis of Malicious Activity," in *2018 Global Information Infrastructure and Networking Symposium (GIIS)*, Thessaloniki, Greece: IEEE, Oct. 2018, pp. 1–4. doi: 10.1109/GIIS.2018.8635778.

[30] V. Mavroeidis and S. Bromander, "Cyber Threat Intelligence Model: An Evaluation of Taxonomies, Sharing Standards, and Ontologies within Cyber Threat Intelligence," in *2017 European Intelligence and Security Informatics Conference (EISIC)*, Athens: IEEE, Sep. 2017, pp. 91–98. doi: 10.1109/EISIC.2017.20.

[31] S. Bistarelli, E. Bosimini, and F. Santini, "A Report on the Security of Home Connections with IoT and Docker Honeypots," p. 12.

[32] C. S. Johnson, M. L. Badger, D. A. Waltermire, J. Snyder, and C. Skorupka, "Guide to Cyber Threat Information Sharing," National Institute of Standards and Technology, NIST SP 800-150, Oct. 2016. doi: 10.6028/NIST.SP.800-150.

[33] P. Há. Meland, K. Bernsmed, E. Wille, Ø. J. Rødseth, and D. A. Nesheim, "A Retrospective Analysis of Maritime Cyber Security Incidents," *TransNav Int. J. Mar. Navig. Saf. Sea Transp.*, vol. 15, no. 3, pp. 519–530, 2021, doi: 10.12716/1001.15.03.04.

[34] K. Tran, S. Keene, E. Fretheim, and M. Tsikerdekis, "Marine Network Protocols and Security Risks," *J. Cybersecurity Priv.*, vol. 1, no. 2, Art. no. 2, Jun. 2021, doi: 10.3390/jcp1020013.

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